CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 3: Query Optimization Strategies & Cost Estimation

Course: Advanced Database Engineering [23CS3503]
Category: Assignments | Assignment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Query processor architecture, relational algebra equivalence rules, and cost metrics.

2. Assignment Problems & Instructions:

1. Transform initial query syntax trees into optimized canonical plans using heuristics.
2. Calculate disk I/O cost estimates for Block Nested Loop Join vs Hash Join vs Merge Join.
3. Explain dynamic programming algorithm used in System R for join order enumeration.
4. Design clustered B+ tree vs non-clustered index strategies for specific query workloads.

Submission Guidelines & Verification:

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